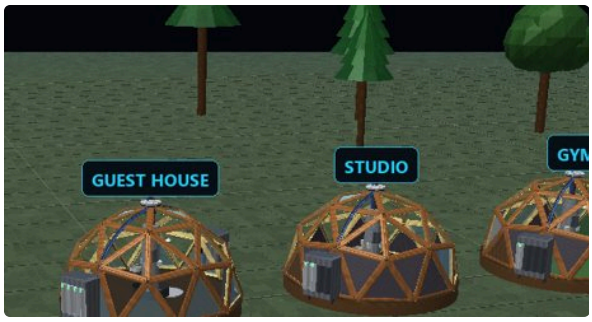
MODULAR HOUSING · BY A VETERAN, FOR VETERANS

The 40 Hour Cabin

One standard body, made the same way every time.

A small home made like a product: one standard frame, a shell in 4 slices, a utility core that carries every service, and snap-in panels that decide what the building is. The standard building lists at $12,036, and every figure on this page is computed by open-source software anyone can run.



$43.45
a sq ft for the standard building, margin in. Published: new house $217, manufactured $87
MODELLED

$1,890
utility core: 5 services, a 25-year life, moves to the next dome for $260
MODELLED

9,870 gal
of rain a year off its own seams, at 40 in of rain
MODELLED

12 + 25
snap-in panel types and modules: a bay changes with a driver, not a builder
MODELLED

15
buildings from one body, $12,036 to $25,511
MODELLED

32,882
veterans homeless on one night in January 2024; 13,851 unsheltered
PUBLISHED (HUD)

THE PROBLEM

Shelter eats the time and money a family needs for everything else: a new site-built house costs **$217 a square foot** before the land, a new manufactured home **$87** (NAHB, Census). On one night in January 2024, **32,882 veterans** had no home of their own (HUD).

THE IDEA: A HOME MADE LIKE A PRODUCT

A manufactured home built as a system, not a one-off. The frame is 40 flat panels on two strut lengths, so every size from 10 to 30 feet uses the same 120 members, the same jig and the same hardware set. The shell comes in 4 slices that two people carry and snap together. Everything that makes it a particular building plugs in.

WHAT PLUGS IN

- **A utility core.** One column carries air, drain, power, structure, water: $1,890, 30% of the build, a 25-year life. Buy the plumbing once.
- **Rain, built in.** The seams between panels are channels: 247 gallons for every inch that falls on 550 sq ft of shell, run to the pad's tank. No gutters.
- **Snap-in panels.** Windows, doors, skylights, solar bays, flues: 12 types and 25 modules, $180 to $1,480.
- **A skin you renew.** Two layers of all-cotton duck soaked in silicone, $1,486 against $2,809 in sailcloth, and a beeswax coat every 1–2 years, about $34 a coat (estimated).

Stem cell · Gym · Studio · Nursery · Guest house · Workshop · Garage · Home · Sauna · Food · Advertiser · Cold storage · Bunker · Treehouse · Jacuzzi: one body, 15 buildings. It stands on a serviced pad and can move: **own the building, rent the ground**.

WHO

Donovan Zeanah. U.S. Navy veteran: six years as a Mass Communication Specialist, three of them in Japan. Learned to program and was certified as an avionics bench technician; now studying for an A&P certificate on the G.I. Bill, on the way to aerospace engineering. Built so far, in the open: the solver, two books (83 and 55 chapters), 34 films and a 6-rung family edition.

WHY I AM DOING THIS

> *Buckminster Fuller wanted housing cheap enough that a mother is not so burdened by it she cannot raise her child — what I took from him, not his words. I grew up with a single parent and a repossessed trailer, and came home from the Navy asking what a small home I owned would have done for my eighteen-year-old self.*

BUILT BY A VETERAN, FOR VETERANS

The life this fits is one a lot of us already know: a small, squared-away space; a system you maintain on a schedule; parts you swap in the field; a crew that builds together. A village of 49 standard buildings on pads, the size of the Veterans Community Project's Kansas City village (VCP), is $0.90 million in buildings and pads before land. It is designed to sit beside HUD-VASH, SSVF, VA Grant and Per Diem, and to be built and kept by veterans.

MEASURED, MODELLED, NOT YET

MEASURED	splitting, 30 wedges per 4-hour session; strut weights; a full-size prototype frame.
MODELLED	cost, energy, rain and the core, in open code.
NOT YET	an engineer's rating; an instrumented year ($12,850); the moulded shell and strut tooling ($60,800).
AGAINST US	53% of the floor has 6.67 ft of headroom; the whole building is about 67 hours, not forty.

THE ASK: $100,650

The sum of its lines, not a round number: prove it, then make it repeatable.

Prove it — A full test platform and one standing dome · Sensors in the wall · An engineer	**$12,850**
Make it repeatable — CNC router · CNC plasma table · Laser cutter · Tooling to mould a strut with its hardware in it · Automated processing for the repetitive work	**$60,800**
Build what an owner cannot — Utility hub · Composite triangle	**$22,000**
Seed the network — Seeding the quilt network	**$5,000**
Total	**$100,650**

Costed as the next step, not asked here: a first site of 3 serviced pads, $100,000.

Every figure computed 13 October 2026 by the project's own software and labelled by kind. Code, books and films: **github.com/dkzeanah/DomeSimulator**

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